

QUESTION-TO-SLIDE MAPPING

SECTION B - LECTURE 04

Classification of Tropical Climate

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	4.0 - Classification of Tropical of Climate (26 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Mapping rule	Map only what the tropical-climate deck actually develops: climatic-zone formation basis, G. A. Atkinson classification, warm-humid equatorial characteristics, hot-dry desert/semi-desert characteristics, and composite/monsoon characteristics. Questions asking passive-control measures, ecological design, or detailed built-form precautions are marked Partial because the deck supplies the climatic problem but not the full design solution.

RESULT

Lecture 04 is a high-return Section B classification deck. It directly supports repeated questions on Atkinson classification, warm-humid climate, hot-dry climate and composite/monsoon climate. Its main limitation is design prescription: the slides are rich in climatic characteristics but do not provide a full passive-control or building-precaution checklist.

Prepared from the uploaded lecture deck and the supplied 2016-2024 regular final question bank.

1. Executive Summary

The lecture begins with how climatic zones arise from solar-atmospheric interaction, gravity and land-sea distribution, then presents Atkinson's 1953 tropical classification based on temperature and humidity extremes affecting comfort. It develops detailed profiles of warm-humid equatorial, hot-dry desert/semi-desert, and composite/monsoon climates through temperature, humidity, precipitation, sky, solar radiation, wind, vegetation and special characteristics. These profiles map strongly to the repeated tropical-climate questions, but design precautions usually require another lecture/source.

26

Slides reviewed

20

Question links

3

Exact matches

8

Direct matches

9

Partial matches

235

Mapped marks

Highest-priority slide blocks

Slides	Lecture block	Historical signals	Exam value
3-5	Atkinson basis + six tropical climate types + world belts	2021 Q2(a); 2022 Q1(c); 2023 Q1(a), Q3(a)	Core classification / zone-formation block.
6-8	Warm-humid equatorial profile	2016 Q2(a); 2021 Q2(b); 2023 Q1(b); 2024 Q2(b)	High-value direct climate-characteristics block.
10-17	Hot-dry desert profile	2016 Q2(a); 2022 Q1(b); 2023 Q1(c); 2024 Q2(c)	High-return contrast + diurnal-range block.
18-25	Composite / monsoon profile	2017 Q2(b); 2021 Q2(c); 2022 Q1(b)	Strong repeated long-answer block.
2	Why climatic zones vary	2022/2023 zone-formation questions	Useful opening mechanism; not a full global-climate derivation.

Priority note

Best order: Slides 3-5 -> 6-8 -> 10-17 -> 18-25. Prepare a 3-column comparison table for warm-humid, hot-dry and composite climates. That single table can be adapted to several repeated questions.

2. Detailed Year-wise Question Mapping

The original printed section label may change from year to year. Mapping below follows subject matter and the lecture content. Slide numbering uses the PDF page count, including the title slide.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2016	Q2(a)	20	1	Compare the characteristics of Hot-dry climate with those of Warm-humid climate.	Direct	6-8, 10-17	Both climate profiles are developed in detail; the student must synthesize them into a comparison table.
2016	Q4(b)	10	1	Explain the major issue of "Ecological Consideration" that can be taken care of by a built-environment designer.	Partial	6-8, 16-17, 24-25	Vegetation, soil, waterlogging, material decay and climatic stresses are described, but no complete ecological-design framework is taught.
2017	Q2(a)	15	2	Explain why passive control in tropical climate is more critical than in sub-tropical or polar climate.	Partial	2-25	The deck shows severe tropical heat/humidity/desert stresses, but it does not teach the passive-control argument or design hierarchy.
2017	Q2(b)	20	2	Characterize the pattern of composite or monsoon climate with example.	Exact	18-25	Location/examples, seasons, temperature, humidity, precipitation, sky, solar radiation, wind, vegetation and material effects are directly covered.
2018	Q5(b)	10	3	What are the major characteristics of tropical climate?	Direct	3-25	The classification and three detailed tropical profiles provide the required characteristic framework.
2018	Q5(c)	10	3	Explain why tropical climates are of maximum importance for built-form designers.	Partial	6-25	The deck shows climate stresses relevant to design, but does not explicitly develop the professional/design-importance argument.
2021	Q2(a)	10	6	Explain the atmospheric factors that affect classification of tropical climates as suggested by G. A. Atkinson.	Exact	3-4	Atkinson, the 1953 classification, and temperature/humidity extremes affecting human comfort are explicitly stated.
2021	Q2(b)	10	6	Describe protective measures that should be taken while constructing buildings in warm-humid climate regions.	Partial	6-8	Warm-humid problems and hazards are described, but protective building measures are not listed in this deck.
2021	Q2(c)	10	6	Characterize composite or monsoon climate with example.	Exact	18-25	Direct lecture match.
2022	Q1(a)	5	8	"Passive control in tropical climate is more critical..." - justify.	Partial	2-25	The climatic need is evident from the profiles; the passive-design justification itself is outside this deck.

2. Detailed Year-wise Question Mapping - continued

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2022	Q1(b)	15	8	Describe the features of composite/monsoon and hot-dry desert climatic patterns with examples.	Direct	10-25	Both requested profiles are directly developed; the answer requires combination/comparison.
2022	Q1(c)	15	8	Explain how climatic zones are formed on Earth and discuss ecological consideration.	Partial	2-5	Slides 2-5 support the broad zone-formation idea; ecological consideration is not developed.
2023	Q1(a)	10	10	Write down the classification of tropical climate and explain how climatic zones are formed on Earth.	Direct	2-5	The six types and broad formation basis are present; the student must expand the mechanism from the concise slide statements.
2023	Q1(b)	10	10	Describe warm-humid equatorial climate with examples.	Direct	5-8	The full climatic profile is present; named city examples are not strongly developed, so the match is Direct rather than Exact.
2023	Q1(c)	10	10	Describe the precautions to be taken while constructing in hot-dry desert climate.	Partial	10-17	The deck explains the climatic hazards and material stress but not a complete construction-precaution checklist.
2023	Q1(d)	5	10	Why is passive control more critical in tropical climate?	Partial	2-25	Climate stress is covered; passive-design reasoning is not.
2023	Q3(a)	10	10	Explain how climatic zones are formed with examples.	Direct	2-5	Solar-atmospheric interaction, gravity and land/sea distribution are given as formation drivers; examples can be drawn from the mapped world belts.
2024	Q2(b)	16	13	Why is the climate warm and humid between the equator and about 15° N/S? Explain using climatic features of the region.	Direct	2, 5-8	Location and warm-humid profile are direct; the deeper global-circulation cause is better supported by Lecture 02.
2024	Q2(c)	12	13	How does desert form in the subtropical region and why is there a large difference between day and night temperatures?	Direct	10-17	Subtropical belt, high diurnal range, clear sky, strong direct solar gain and rapid night re-radiation are all present.
2024	Q8(a)	12	15	Identify the climates represented by two traditional built forms and analyse components, surroundings and orientation.	Partial	6-8, 10-17	The deck can identify warm-humid vs hot-dry climatic signatures, but built-form component/orientation analysis requires design lectures.

Coverage key

Exact = answer content is essentially present as asked. Direct = most content is present, but the student must apply/combine it. Partial = only one component is present; another lecture/source is required.

3. Repeated Question Patterns

The same core task repeats in different wording: identify a tropical climate, describe its climatic profile, then relate it to design. The lecture is strongest for the first two steps. When the question asks “what should the building do?”, switch to the climate-responsive design / microclimate lectures for the design prescription.

P	Normalized repeated family	Years seen	Slides	Why it matters
S	Warm-humid vs hot-dry characteristics / contrast	2016, 2021, 2022, 2023, 2024	6-17	Most reusable profile pair; table answer works well.
S	Composite / monsoon climate characteristics	2017, 2021, 2022	18-25	Repeated long-answer family with explicit examples and seasonal data.
S	Tropical classification / climatic-zone formation	2021, 2022, 2023	2-5	Core classification logic; often 10-15 marks.
A	Warm-humid equatorial climate	2021, 2023, 2024	5-8	Recent direct questions; memorize data ranges and hazards.
A	Hot-dry desert formation / large diurnal range	2022, 2023, 2024	10-17	Recent direct + application family.
B	Passive-control / construction precautions	2017, 2021-2023	Climate profiles only	Important questions, but this lecture is only a partial source.

4. Quick Recognition Rules

If the question says...	Go to slides	Recognition rule
“Atkinson / classify tropical climates”	3-4	Basis = extremes of temperature + humidity affecting human comfort; six named types.
“warm-humid equatorial”	5-8	Near ~15° N/S; high RH; heavy rainfall; cloudy sky; low wind; rapid vegetation/mould/rot.
“hot-dry desert / semi-desert”	10-17	~15-30° N/S; very high day temperature; large diurnal range; low RH/rain; clear sky; dusty hot winds.
“composite / monsoon”	18-25	Seasonal mix: hot-dry + warm-humid (+ sometimes cool-dry); monsoon rain and seasonal wind/sky changes.
“compare warm-humid vs hot-dry”	6-8 + 12-17	Use one row each for temperature, RH, rainfall, sky/solar, wind, vegetation, design problem.
“protective measures / passive control”	Other design lectures	Use this deck for climatic problem only; do not invent the design solution from profile data.

5. Slide-to-Question Reverse Index

Use this page while revising the lecture: start from a slide block, then immediately practice the linked past questions.

Slides	What the slides teach	Past-question links	Study action
2-5	Climatic-zone formation + Atkinson classification + world belts	2021 Q2(a); 2022 Q1(c); 2023 Q1(a), Q3(a)	Memorize formation sentence + six-type tree.
6-8	Warm-humid equatorial climate profile	2016 Q2(a); 2021 Q2(b) partial; 2023 Q1(b); 2024 Q2(b)	Make a 7-row climate-data table.
10-17	Hot-dry desert/semi-desert profile	2016 Q2(a); 2022 Q1(b); 2023 Q1(c) partial; 2024 Q2(c)	Focus on diurnal range, clear sky, strong solar, low RH/rain and dust.
18-25	Composite / monsoon profile	2017 Q2(b); 2021 Q2(c); 2022 Q1(b)	Prepare as seasonal comparison: hot-dry / warm-humid / cool-dry.
26	Closing	None	Skip.

6. Coverage Gaps & Diagnostic Exclusions

The lecture is descriptive rather than prescriptive. It tells you what each tropical climate is like, but many exam questions add a second demand: “what should a designer do about it?” That second part is not fully answered here.

Gap	Affected questions	Action
Passive-control rationale is not developed	2017 Q2(a); 2022 Q1(a); 2023 Q1(d)	Use the introductory/passive-design lecture and climate-responsive design notes.
Warm-humid / hot-dry building precautions are not listed	2021 Q2(b); 2023 Q1(c)	Use Design Specifications / Microclimate / Envelope lectures.
Ecological consideration is not developed as a design framework	2016 Q4(b); 2022 Q1(c)	Use the relevant class note/reference; keep climate-profile facts separate.
Traditional built-form component/orientation analysis is absent	2024 Q8(a)	Use the question-bank solution/design lectures; Lecture 04 only helps identify the climate.
Warm-humid “why near equator?” global mechanism is concise	2024 Q2(b)	Pair with Lecture 02 global solar/wind/ITCZ material for a deeper causal answer.

Mapping discipline

All numerical climate ranges in this report are treated as course-slide values. The report does not silently replace or normalize them with external climatology. Where a question asks design measures not printed in the deck, the mapping stays Partial.

7. Recommended Study Plan

Do not memorize 26 slides as isolated facts. Build three master climate profiles and one classification tree. Then practise converting a profile into a comparison answer.

Step	Study block	Slides	Practice immediately
1	Classification tree + Atkinson basis	2-5	2021 Q2(a), 2023 Q1(a).
2	Warm-humid profile	6-8	2023 Q1(b), 2024 Q2(b).
3	Hot-dry profile	10-17	2016 Q2(a), 2024 Q2(c).
4	Composite / monsoon profile	18-25	2017 Q2(b), 2021 Q2(c), 2022 Q1(b).
5	Design-gap check	All	For any "measures/precautions/passive" wording, open the design lectures before writing.

Exam-ready minimum

Priority	What to reproduce	Exam technique
A	Six-type classification tree	Write Atkinson + temperature/humidity basis above the tree.
A	Warm-humid vs hot-dry table	At least 7 comparison rows + one-line design implication.
A	Composite seasonal table	Hot-dry / warm-humid / cool-dry temperature + RH + rain + wind/sky.
B	World-belt sketch	Mark equator, ~15° and ~30° N/S and show warm-humid / hot-dry belts.

Bottom line

Lecture 04 is a must-prepare descriptive climate lecture. Master the three climate profiles first; for any question that adds “design measures”, treat this deck as the problem-definition source and combine it with the design lectures.